Credit Card Approval Policy: Profit Optimization

Assumptions: revenue 6% and loss 30% of credit limit (placeholders). Data: UCI Taiwan credit card default, 30,000 customers.

Customers analysed	Overall default rate	Profit: approve everyone (NT$)	Profit: recommended policy (NT$)	Improvement (NT$)	Approval rate (recommended)
30000	**22%**	**0.15M**	**25.85M**	**25.70M**	**68%**

Recommendation

1. Approve customers who are not late on payments and have credit limits of 50,000 or more (S2, S3): 68% of applicants.
2. Decline or manually review customers who are 1+ months late: their Test default rates run from 25% to 75%.
3. Also approving the small-limit, not-late group (S1) lifts approvals to 78% and costs about 1% of profit.
4. S3 earns about 95% of the profit. S2 is marginal: it stops paying off if the loss per default exceeds about 5.5x the revenue per good customer (base case: 5x).
5. All figures rest on placeholder revenue and loss rates and one 70/30 split; replace them with the bank's own figures before real use.

Chart 1: Default rate by segment, Train vs Test:

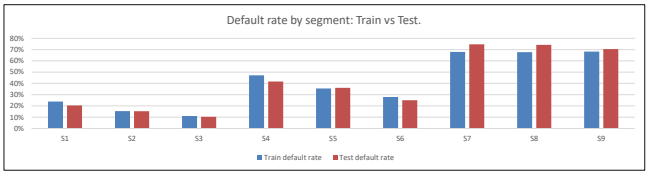


Chart 2: Actual profit by segment

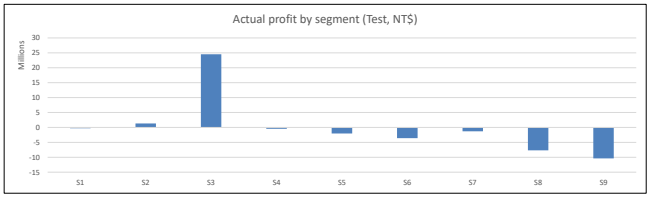
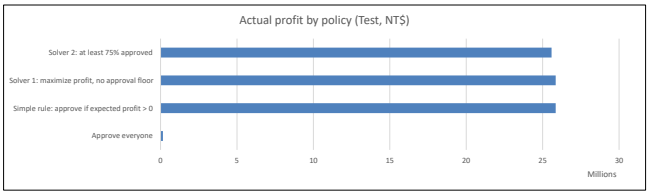


Chart 3: Policy comparison



Approval rate under different assumptions

Loss (down) / Revenue (across)	4%	5%	6%	7%	8%
20%	68%	68%	68%	78%	83%
25%	41%	68%	68%	68%	78%
30%	41%	41%	68%	68%	68%
35%	0%	41%	41%	68%	68%
40%	0%	41%	41%	41%	68%
50%	0%	0%	0%	41%	41%

Rows: loss per default (% of credit limit). Columns: revenue per good customer. Base case outlined: 6% revenue, 30% loss. Green = more applicants approved, red = fewer.